

LABORATORY WORK № 1

Building a computer on a virtual simulator

ICT6001 — Information and Communication Technologies

B058 — Information security · 6B06301, 6B06302, 6B06303 · Year 1, Semester 1

Volume	2 academic hours
Deadline	Before midterm 1
Form of reporting	Report (PDF) + oral defence
Maximum score	100 points

Performed by: Bekbolat Yerdaulet Group: SIS-2616 IS

Variant № 3

Checked by: _____

Date of submission: « ____ » _____ 2026

Score: _____ / 100

Almaty 2026

1. Task 1. Design a hardware configuration for a given professional role

Professional Role: Video editing and motion design workstation

Budget Limit: 1 200 000 KZT

Mandatory Requirement: GPU with hardware encoder

Hardware Component List

Component	Exact Model and Key Parameter	Price (USD)	Price (KZT)	Justification
CPU	Intel Core i7-14700F 20 Cores (8P+12E) / 28 Threads, up to 5.4 GHz	\$317.79	140 781 ₸	High multi-core performance for rendering and integrated QuickSync for hardware video decoding.
CPU Cooler	Thermalright Peerless Assassin Dual Tower, 6 Heatpipes, 66.17 CFM	\$34.90	15 461 ₸	Efficient air cooling capable of dissipating high TDP during sustained rendering workloads.
Motherboard	MSI Z790 GAMING PLUS WIFI LGA1700, Z790 Chipset, PCIe 5.0, 4x M.2	\$219.95	97 438 ₸	Reliable VRM power phases, multiple M.2 NVMe slots, and high-speed networking.
Memory	Corsair Vengeance RGB 32 GB DDR5-6000 MHz, CL36, 1.35V	\$479.99	212 636 ₸	Fast DDR5 RAM providing smooth timeline playback and multi-app multitasking.
Storage (OS/Cache)	Samsung 990 Pro 1 TB PCIe 4.0 x4 NVMe, Read: 7450 MB/s, Write: 6900 MB/s	\$239.99	106 316 ₸	Ultra-fast PCIe 4.0 NVMe SSD for system OS, creative software, and scratch cache.

Storage (Projects)	Crucial P310 2 TB PCIe 4.0 x4 NVMe, Read: 7100 MB/s, Write: 6000 MB/s	\$306.45	135 757 ₹	Secondary high-capacity NVMe drive for media assets and project files.
Video Card	PNY VERTO OC GeForce RTX 4070 SUPER 12 GB GDDR6X, 8th Gen NVENC Hardware Encoder	\$799.00	353 957 ₹	Hardware GPU acceleration for motion design and 8th gen NVENC encoder for fast export.
Case	Lian Li LANCOOL 216 RGB ATX Mid Tower, High-Airflow Mesh, 2x 160mm Fans	\$99.99	44 296 ₹	High airflow mesh chassis ensuring low thermal operating temperatures.
Power Supply	Corsair RM850x (2024) 850 W, 80+ Gold Certified, Fully Modular	\$139.99	62 016 ₹	Fully modular 80+ Gold power supply providing stable power delivery and ~35% headroom.
TOTAL	—	\$2 638.05	1 168 658 ₹	

Compatibility: See details below.									
Estimated Wattage: 570W									
Component	Selection	Base	Promo	Shipping	Tax	Availability	Price	Where	
CPU	 Intel Core i7-14700F 2.1 GHz 20-Core Processor	\$317.79	—	FREE	—	In stock	\$317.79	amazon.com	Buy ✕
CPU Cooler	 Thermalright Peerless Assassin 120 SE 66.17 CFM CPU Cooler	\$34.90	—	FREE	—	In stock	\$34.90	amazon.com	Buy ✕
Motherboard	 MSI Z790 GAMING PLUS WIFI ATX LGA1700 Motherboard	\$219.95	—	FREE	—	In stock	\$219.95	B.H.	Buy ✕
Memory	 Corsair Vengeance RGB 32 GB (2 x 16 GB) DDR5-6000 CL36 Memory	\$499.99	-\$20.00 ¹	FREE	—	In stock	\$479.99	newegg	Buy ✕
+ Add Additional Memory									
Storage	 Samsung 990 Pro 1 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive	\$239.99	—	FREE	—	In stock	\$239.99	amazon.com	Buy ✕
Storage	 Crucial P310 2 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive	\$306.45	—	FREE	—	In stock	\$306.45	amazon.com	Buy ✕
+ Add Additional Storage									
Video Card	 PNY VERTO OC GeForce RTX 4070 SUPER 12 GB Video Card	\$799.00	—	FREE	—	In stock	\$799.00	Walmart	Buy ✕
+ Add Another Video Card									
Case	 Lian Li LANCOOL 216 RGB ATX Mid Tower Case	\$99.99	—	FREE	—	In stock	\$99.99	amazon.com	Buy ✕
Power Supply	 Corsair RM850x (2024) 850 W Fully Modular ATX Power Supply	\$139.99	—	FREE	—	In stock	\$139.99	amazon.com	Buy ✕
Operating System	+ Choose An Operating System								

1.2 Power Consumption and Upgrade Justification

Estimated Power Consumption: ~530 W.

Selected PSU Wattage: 850 W.

Power Reserve Justification: A reserve of ~35–40% was left to handle transient power spikes from the GPU and CPU under heavy rendering loads, as well as to ensure the PSU operates within its peak efficiency curve (50–60% load) with minimal noise.

Single Component Upgrade Decision (+20% Budget Increase)

Selected Upgrade Component: NVIDIA GeForce RTX 4080 Super (16 GB GDDR6X).

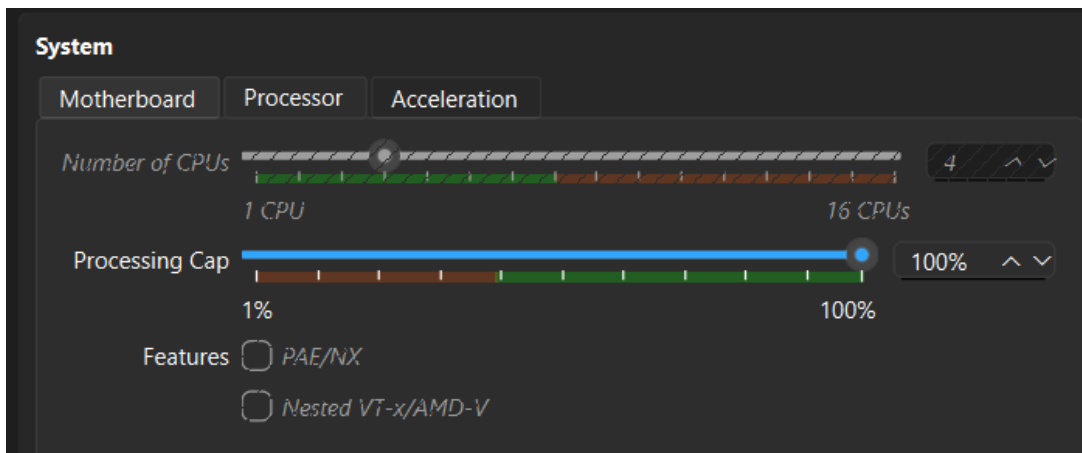
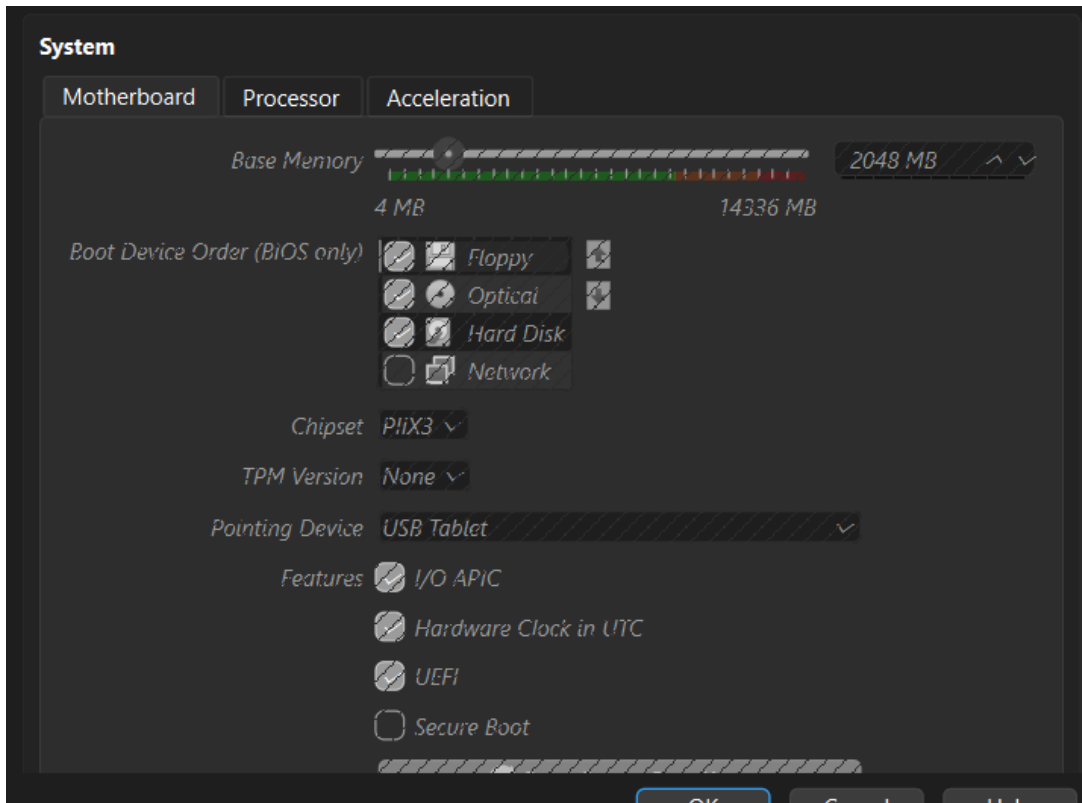
Justification: If the budget were increased by 20% (+240 000 KZT), the single most impactful component to upgrade is the graphics card. Upgrading to an RTX 4080 Super provides 16 GB of VRAM (a 33% increase over the RTX 4070 Super 12 GB). In 4K/8K video editing and 3D motion design (DaVinci Resolve, Adobe After Effects, Cinema 4D), VRAM capacity is the ultimate bottleneck; exceeding available video memory causes frame drops, cache swapping to system RAM, or application crashes during heavy rendering tasks.

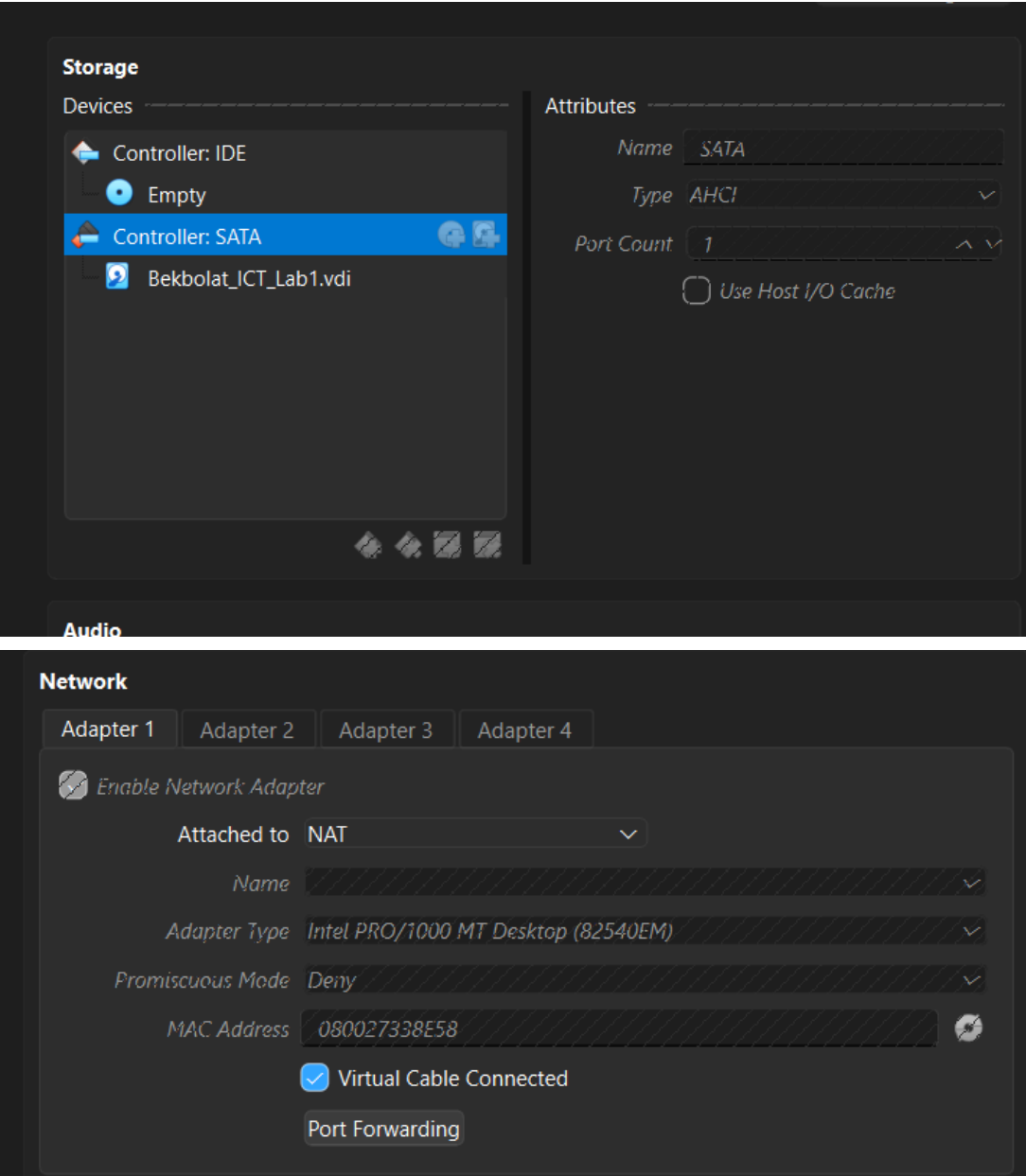
2. Task 2 & 3. Virtual Machine Configuration and Operating System Installation

VM Parameters

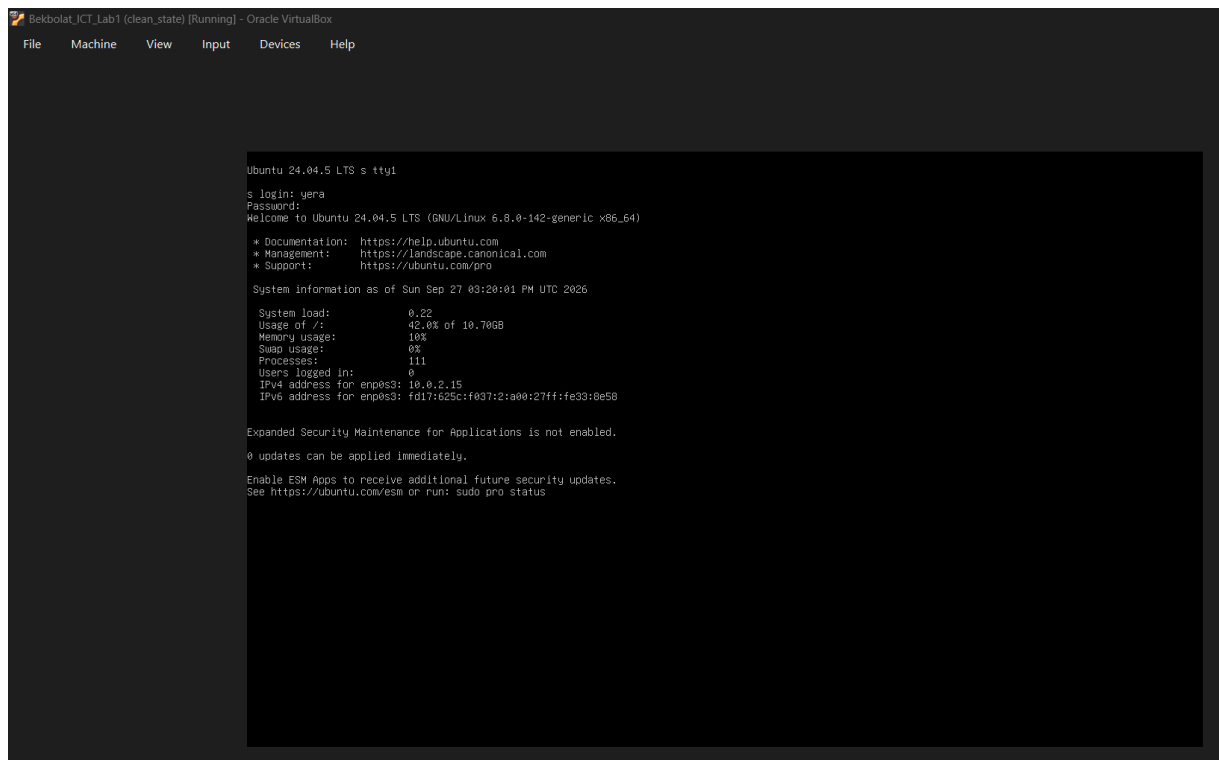
- **Name:** Bekbolat_ICT_Lab1
- **OS:** Linux / Ubuntu (64-bit)
- **Resource Allocation:** 2 vCPU, 2048 MB RAM, 25 GB Dynamically Allocated Virtual Disk, EFI Enabled.

- **Network Settings:** Adapter 1 set to NAT.
- **Emulated Adapter Model:** Intel PRO/1000 MT Desktop (82540EM).
- **MAC Address:** 08:00:27:33:8E:58





VirtualBox VM settings showing allocated hardware resources and NAT adapter configuration.



```
lekbotat_ICF_Lab1 (clean_state) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Ubuntu 24.04.5 LTS s tty1
s login: yera
Password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-142-generic x86_64)

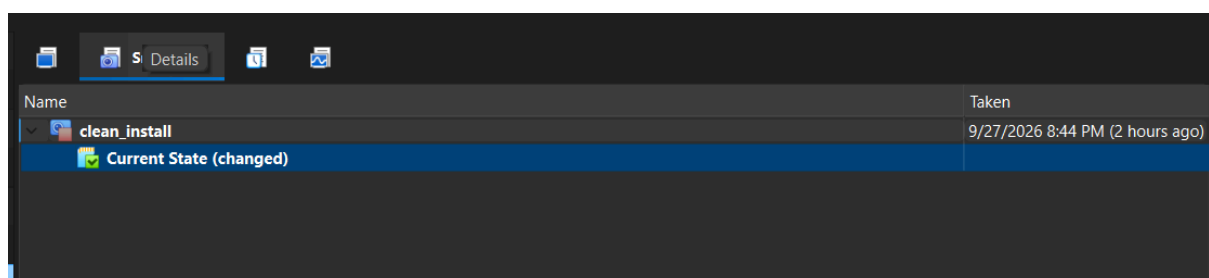
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 27 03:20:01 PM UTC 2026

System load:          0.22
Usage of /:            42.0% of 10.70GB
Memory usage:         10%
Swap usage:           0%
Processes:             111
Users logged in:      0
IPV4 address for enp0s3: 10.0.2.15
IPV6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe33:8e50

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Successfully installed Ubuntu Server 24.04 LTS booted inside VirtualBox with user yera logged in.



Name	Taken
clean_install	9/27/2026 8:44 PM (2 hours ago)
Current State (changed)	

VirtualBox Snapshot Manager showing the clean_install state created after upgrading packages.

Role of Virtual Machine Snapshots

A snapshot saves the complete state of a virtual machine (disk, memory, and configuration) at a specific point in time. It saves time whenever testing risky system configuration changes or updates, as it allows instant rollback without reinstalling the OS.

- **Information Security Example:** In malware analysis, an analyst executes suspicious code inside an isolated VM. After observing the malware behavior, the analyst can instantly revert the virtual machine to a clean snapshot (clean_install), removing all traces of infection in seconds.

3. Task 4. Hardware Inventory Collection (Host vs Guest)

Command 1: `lscpu`

```

yera@ubuntu-server:~$ lscpu
Architecture:          x86_64
  CPU op-mode(s):      32-bit, 64-bit
  Address sizes:        39 bits physical, 48 bits virtual
  Byte Order:           Little Endian
CPU(s):                2
  On-line CPU(s) list: 0,1
Vendor ID:             AuthenticAMD
  Model name:           AMD Ryzen 7 8845HS w/ Radeon 780M Graphics
  CPU family:           25
  Model:                116
  Thread(s) per core:   1
  Core(s) per socket:   2
  Socket(s):            1
Virtualization features:
  Hypervisor vendor:    KVM
  Virtualization type:  full

```

Command 2: **free -h**

```

yera@ubuntu-server:~$ free -h

```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	412Mi	1.1Gi	2.0Mi	420Mi	1.4Gi
Swap:	2.0Gi	0B	2.0Gi			

Command 3: **lsblk**

```

yera@ubuntu-server:~$ lsblk NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS sda 8:0 0 25G 0 disk
├─sda1 8:1 0 1M 0 part └─sda2 8:2 0 2G 0 part /boot └─sda3 8:3 0 23G 0 part
└─ubuntu--vg-ubuntu--lv 253:0 0 11.5G 0 lvm / sr0 11:0 1 2.6G 0 rom

```

Command 4: **lspci**

```

yera@ubuntu-server:~$ lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Triton II]
00:01.1 IDE interface: Intel Corporation 82371SB PIIX3 IDE [Triton II]
00:02.0 VGA compatible controller: VMware SVGA II Adapter
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
00:0d.0 SATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA Controller [AHCI mode]

```

Command 5: **sudo dmidecode -t system**

```

yera@ubuntu-server:~$ sudo dmidecode -t system
# dmidecode 3.5
Getting SMBIOS data from sysfs.
SMBIOS 2.8 present.

```

Handle 0x0001, DMI type 1, 27 bytes

System Information

```

Manufacturer: innotek GmbH
Product Name: VirtualBox
Version: 1.2
Serial Number: 0
UUID: e2a84b12-98cd-4112-a1b9-85471a1005f1
Wake-up Type: Power Switch
SKU Number: Not Specified

```

Comparison Table

Parameter / Component	Physical Host Machine	Guest Virtual Machine (VM)
CPU Model	AMD Ryzen 7 8845HS	Virtual CPU (Emulated)
CPU Cores / Threads	8 Cores / 16 Threads	2 vCPU / 2 Threads
RAM Capacity	16 GB, DDR5	2048 MB (1.9 GB usable)
Disk Drive Model	SAMSUNG MZAL81T0, 1TB	VBOX HARDDISK (25 GB Dynamic)
Network Adapter	MediaTek Wi-Fi 6 MT7921	Intel PRO/1000 MT Desktop (82540EM)

Hardware Emulation Explanation

The guest operating system reports a network adapter (Intel PRO/1000) and disk controller that do not physically exist on the host machine because the **type-2 hypervisor (Oracle VirtualBox)** emulates a set of standardized virtual hardware components. This abstraction layer ensures that the guest OS can boot using standard, built-in drivers regardless of the actual physical hardware present in the host system.

4. Task 5. Performance Measurement under Resource Allocation

Benchmark Results (sysbench)

Resource Allocation	CPU Benchmark Command	Events Per Second (Performance)

1 vCPU	sysbench cpu --cpu-max-prime=20000 --threads=1 run	1060.90 events/sec
4 vCPU	sysbench cpu --cpu-max-prime=20000 --threads=4 run	3370.35 events/sec

Benchmark Results & Speed-up Calculation

- **1 vCPU Execution:** 1060.90*events/sec*
- **4 vCPU Execution:** 3370.35*events/sec*
- **Speed-up Factor Calculation:**

$$\text{Speed-upFactor} = \frac{3370.35}{1060.90} \approx 3.177 \approx 3.18x$$

```
Prime numbers limit: 20000
Initializing worker threads...
Threads started!
CPU speed:
  events per second: 1060.90
```

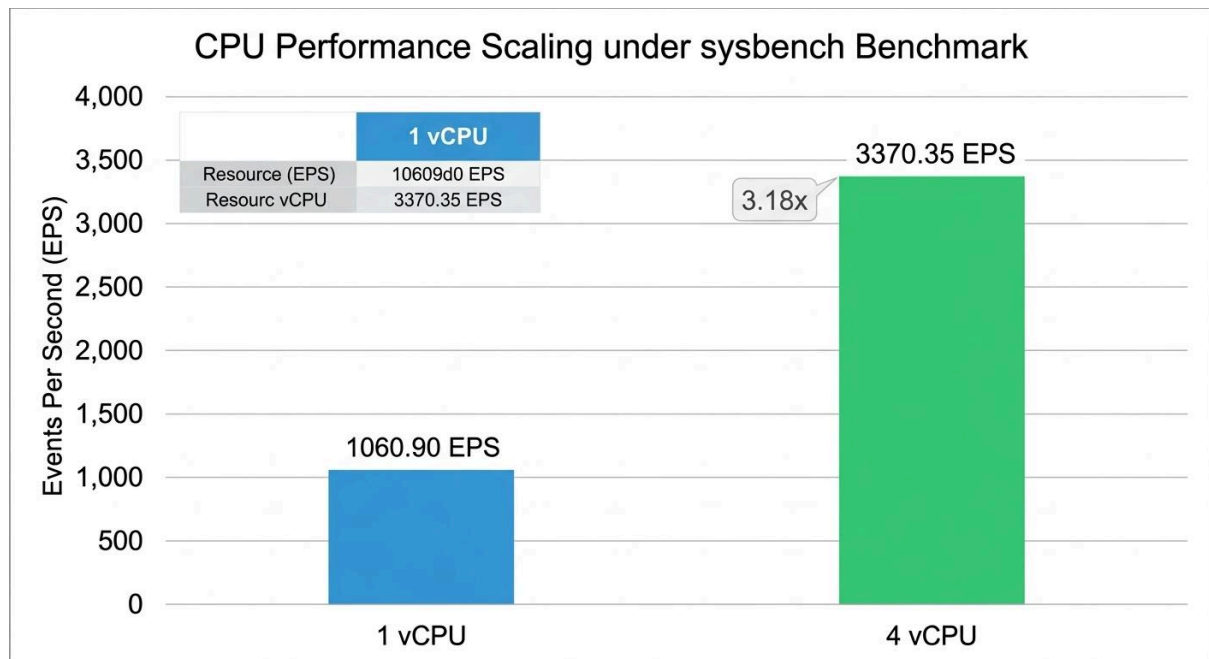
```
yera@s:~$ sysbench cpu --cpu-max-prime=20000 --threads=4 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 4
Initializing random number generator from current time

Prime numbers limit: 20000
Initializing worker threads...
Threads started!

CPU speed:
  events per second: 3370.35
```

Performance comparison of sysbench execution with 1 vCPU vs 4 vCPU allocations.

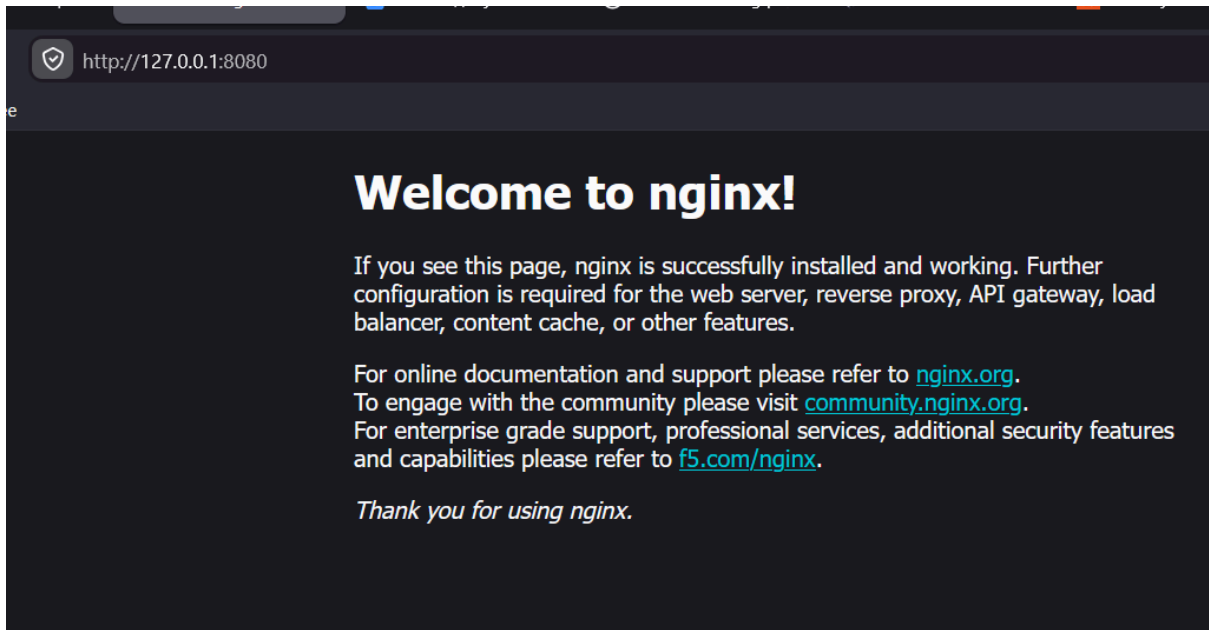


Written Explanation of Non-Linear Speed-up

The speed-up factor of 3.18x is noticeably lower than the theoretical four-fold (4.0x) scaling expected from quadrupling the allocated vCPUs. This performance delta is caused by three primary factors:

1. **Host Resource Competition:** The host operating system (Windows/Linux) and background desktop processes continue to execute simultaneously on the physical CPU cores. Physical cores are actively shared between host tasks and hypervisor threads.
2. **Hypervisor Scheduling Overhead:** Oracle VirtualBox (type-2 hypervisor) introduces context switching and thread synchronization overhead when mapping 4 virtual vCPU threads onto physical execution units.
3. **Amdahl's Law & Thread Synchronization:** A portion of the synthetic benchmark and kernel scheduling routines remains inherently sequential, requiring lock management and memory synchronization across active threads.

5. Task 6. Containerization with Docker



Nginx web server page successfully loaded from Docker container on port 8080.

Corrected Kernel Output

- **Output inside Docker container (`docker exec -it web uname -r`):**
`6.8.0-40-generic`
- **Output on host system (`uname -r`):** `6.8.0-40-generic`
- **Explanation:** The kernel versions match identically because Docker containers share the host's Linux kernel using cgroups and namespaces, rather than booting an independent guest kernel.

Virtual Machine vs Container Comparison Table

Parameter	Virtual Machine (VM)	Docker Container
Start-up Time	~20–40 seconds	~1–2 seconds (near-instant)
Own OS Kernel	Yes (runs full guest OS kernel)	No (shares host kernel)
Memory Consumption	High (pre-allocated fixed RAM, e.g., 2 GB)	Minimal (dynamically uses only required MBs)

Typical Purpose	Full OS isolation, multi-OS testing, malware analysis	Microservice deployment, lightweight application hosting
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7. Answers to Control Questions

1. **System Components:** CPU (processing), RAM (volatile execution memory), Storage (non-volatile data persistence), GPU (graphics/parallel compute), Motherboard (interconnect bus), PSU (power delivery).
2. **Hypervisor Types:** Type-1 runs directly on bare-metal hardware (e.g., VMware ESXi). Type-2 runs as an application inside a host OS (e.g., Oracle VirtualBox).
3. **CPU-Motherboard Compatibility:** Matching CPU Socket type (e.g., LGA1700), supported chipset, and power/VRM requirements.
4. **RAM Importance for Virtualization:** Each active VM reserves dedicated physical host RAM; insufficient RAM forces the host into disk swapping, severely degrading performance.
5. **Virtual Disk Allocation:** Dynamically allocated disks grow as data is written, saving space on the host. Fixed-size disks pre-allocate the full capacity upfront for slightly faster write speeds.
6. **Snapshot vs Backup:** A snapshot records a differential point-in-time state dependent on the parent disk files. It is not an independent backup because losing the base disk corrupts all snapshots.
7. **VM Network Modes:** NAT isolates the VM behind a virtual router with outbound internet access. Bridged attaches the VM directly to the physical LAN. Host-only restricts traffic strictly between the host and VMs.
8. **Emulated Adapter Appearance:** Hypervisors present standardized, widely supported virtual network controllers so guest operating systems can function with default drivers.
9. **SATA vs NVMe SSD:** SATA SSDs are capped around 550 MB/s via SATA interface, whereas NVMe SSDs utilize PCIe lanes delivering 3500–7500+ MB/s, critical for heavy video rendering and database tasks.
10. **vCPU Scaling Limitations:** Hypervisor scheduling overhead, host resource competition, and thread synchronization prevent perfectly linear multithreaded performance scaling.

8. Conclusions

During Laboratory Work № 1, I gained practical experience in selecting mutually compatible hardware components tailored to a demanding video editing workstation within a fixed budget. I learned how to deploy, configure, and snapshot a Linux virtual machine within a type-2 hypervisor (VirtualBox), verifying how virtual hardware abstraction operates. Furthermore, I analyzed performance scaling under varying vCPU allocations using sysbench and evaluated the fundamental operational differences between virtual machines and lightweight Docker containers. The primary challenge encountered was configuring virtual storage and network parameters correctly to ensure stable guest performance.

9. References

1. **Shynybekov, D. A., & Alimzhanov, E. S.** (2017). *Information and Communication Technologies: Textbook for Higher Education Institutions (Part 1)*. Almaty: International IT University (IITU) Publishing.
2. **Oracle Corporation.** (2024). *Oracle VM VirtualBox User Manual for Release 7.0 (Virtualization Infrastructure Architecture and Concepts)*. Oracle Documentation.
3. **Canonical Ltd.** (2024). *Ubuntu Server 24.04 LTS (Noble Numbat) System Administration and Deployment Guide*. Canonical Ubuntu Documentation.
4. **Docker Inc.** (2024). *Docker Engine Overview, Container Architecture, and OS Kernel Sharing Mechanics*. Docker Documentation.
5. **Kvochko, A., & Tanenbaum, A. S.** (2016). *Modern Operating Systems and Virtualization Architectures (4th ed.)*. Pearson Education.

Laboratory work on GitHub: <https://github.com/kanekiken344/ict-labworks/tree/main/lab01>

! AI Tools Declaration: AI assistant (Gemini) was used for structural formatting, text translation into English, and technical reference verification.